

PayStone Customer Retention Intelligence Platform: AI-Based Churn Prediction and Explainable Analytics

Business Overview

PayStone is a Canadian fintech company that provides innovative digital lending and financial services to individuals and small businesses across Canada. By leveraging artificial intelligence (AI), cloud computing, and advanced analytics, PayStone delivers fast, secure, and customer-centric financial solutions through its digital platform. Customers can conveniently apply for loans, manage their accounts, make repayments, and access personalised financial products without visiting a physical branch.

As the company has grown rapidly in Canada's highly competitive fintech market, PayStone has focused on enhancing customer experience through personalised services, digital engagement, and data-driven decision-making. Maintaining long-term customer relationships has become a strategic priority, as customer retention is significantly more cost-effective than acquiring new customers.

Business Problem

As PayStone continues to expand its customer base, **customer churn has emerged as a major business challenge**. An increasing number of customers are discontinuing their relationship with the company by closing their accounts, moving to competing financial providers, or becoming inactive. This trend negatively impacts revenue growth, customer lifetime value, and long-term business sustainability.

Customer churn is influenced by multiple factors, including declining customer engagement, dissatisfaction with services, limited product usage, poor customer support experiences, competitive offers from rival fintech companies, and changing financial needs. Identifying customers who are likely to leave is difficult using traditional reporting methods because churn is driven by complex interactions between demographic, behavioural, transactional, and service-related variables.

High churn rates create several business challenges, including:

- Reduced recurring revenue and customer lifetime value (CLV).
- Increased customer acquisition costs to replace lost customers.
- Lower profitability due to declining customer retention.
- Reduced cross-selling and upselling opportunities.
- Loss of market share in Canada's competitive fintech industry.
- Inefficient allocation of marketing and customer retention resources.

To address these challenges, PayStone aims to develop an **Explainable Artificial Intelligence (XAI)-driven Customer Churn Prediction System** capable of identifying customers who are at high risk of leaving before they churn. The solution leverages advanced machine learning algorithms to analyse customer demographics, account activity, transaction behaviour, product usage, engagement metrics, and service interactions to predict churn probability.

To ensure transparency and trust in AI-driven decision-making, the predictive model incorporates **SHAP (SHapley Additive Explanations)**, enabling business stakeholders to understand the key factors influencing each prediction. This explainability supports data-driven retention strategies while promoting responsible and transparent AI adoption.

The proposed solution will enable PayStone to:

- Predict customers who are most likely to churn.
- Proactively target at-risk customers with personalised retention campaigns.
- Improve customer satisfaction and loyalty through timely interventions.
- Increase customer lifetime value and long-term profitability.
- Optimise marketing expenditure by focusing retention efforts on high-risk customers.
- Provide transparent and explainable AI insights to support business decision-making.

By adopting Explainable AI for customer churn prediction, PayStone can strengthen customer relationships, improve retention rates, and gain a competitive advantage in Canada's rapidly evolving fintech sector while making more informed, transparent, and data-driven business decisions.

For a fintech portfolio project, **Render** is the strongest choice because it supports production-style deployment of Python APIs (Flask/FastAPI), is free for portfolio use, and closely resembles how machine learning models are operationalised in industry.

Here is the revised **Project Scope** with the deployment included:

Project Scope

This project analyses **PayStone's customer data from January 2024 onwards** to develop and evaluate an **AI-driven Customer Churn Prediction model**. The dataset includes customer demographics, loan information, repayment behaviour, account activity, transaction history, digital platform usage, customer service interactions, and customer engagement metrics.

The collected data are used to perform data preprocessing, feature engineering, exploratory data analysis, and predictive modelling using advanced machine learning algorithms. Model performance is evaluated using appropriate classification metrics to identify the most effective model for predicting customer churn. To enhance transparency and build stakeholder trust, **Explainable Artificial Intelligence (XAI)** techniques, including **SHAP (SHapley Additive Explanations)**, are incorporated to interpret model predictions and identify the key factors influencing customer churn.

The trained machine learning model is deployed as a cloud-hosted prediction service using **FastAPI** on **Render**, enabling secure and scalable access to real-time churn predictions through a RESTful API. This deployment approach reflects modern industry practices for operationalising machine learning models within fintech environments and supports seamless integration with downstream business applications.

To support operational and strategic decision-making, the project includes the development of an interactive **Power BI monitoring dashboard** that visualises customer churn trends, prediction outcomes, model performance, key churn drivers, customer risk profiles, and business KPIs. The dashboard provides business stakeholders with real-time insights to monitor customer churn risk, evaluate retention strategies, and support data-driven decision-making. Overall, the project demonstrates how artificial intelligence, explainable machine learning, cloud-based model deployment, and business intelligence can be integrated to improve customer relationship management, reduce customer attrition, and enhance long-term business performance within the Canadian fintech lending industry.

6. Machine Learning Modelling Approach

A range of supervised machine learning classification algorithms will be developed, trained, and evaluated to identify the most effective model for predicting customer churn within the PayStone fintech lending environment.

The modelling approach follows an incremental methodology, beginning with an interpretable statistical baseline model and progressing towards advanced ensemble learning techniques. This allows the project to compare model performance, evaluate trade-offs between interpretability and predictive accuracy, and select the most suitable algorithm for deployment. The following machine learning models will be implemented:

Baseline Model

Logistic Regression

- Logistic Regression will be developed as the baseline classification model for customer churn prediction.
- The model estimates the probability of a customer churning based on the relationship between customer characteristics and behavioural variables.
- It provides a highly interpretable benchmark, allowing business stakeholders to understand the direction and impact of individual features on churn probability.
- Logistic Regression is widely used within financial services due to its transparency, simplicity, regulatory acceptance, and suitability for risk-related decision-making.

Traditional Machine Learning Models

Decision Tree



- Decision Tree models will be implemented to capture complex, non-linear relationships between customer characteristics and churn outcomes.
- The algorithm creates a sequence of decision rules based on customer attributes, making predictions through a tree-like structure.
- Decision Trees provide intuitive explanations of customer churn patterns, allowing stakeholders to understand the conditions that contribute to customer attrition.
- However, additional regularisation techniques may be required to prevent overfitting when modelling complex customer behaviour.

Random Forest

- Random Forest will be implemented as an ensemble learning technique that combines multiple decision trees to improve prediction accuracy and model stability.
- By aggregating predictions from multiple trees, the algorithm reduces the risk of overfitting and improves generalisation performance on unseen customer data.
- Random Forest can effectively capture interactions between demographic, financial, transactional, and engagement-related features.
- The model also provides feature importance rankings, helping identify the customer characteristics that have the greatest influence on churn prediction.

Advanced Ensemble Models

XGBoost (Extreme Gradient Boosting)

- XGBoost will be implemented as an advanced gradient boosting algorithm designed to achieve high predictive performance on structured customer datasets.
- The model builds multiple decision trees sequentially, where each new tree improves the weaknesses of previous models by reducing prediction errors.
- XGBoost is widely adopted across banking and fintech organisations due to its ability to capture complex relationships, handle nonlinear customer behaviour patterns, and provide strong predictive accuracy.
- The algorithm is expected to perform effectively on PayStone's customer churn dataset containing demographic, transaction, engagement, and financial behaviour variables.

LightGBM (Light Gradient Boosting Machine)

- LightGBM will be evaluated as an efficient gradient boosting framework designed for high-performance machine learning applications.
- The algorithm uses a leaf-wise tree growth approach, allowing it to achieve strong predictive performance while maintaining computational efficiency.



- LightGBM is particularly suitable for large-scale customer analytics applications where organisations need fast model training and scalable prediction capabilities.
- The model can effectively identify subtle patterns in customer behaviour that may indicate future churn risk.

CatBoost (Categorical Boosting)

- CatBoost will be implemented due to its ability to handle categorical customer variables effectively with minimal preprocessing.
- The algorithm is particularly suitable for fintech customer analytics datasets containing variables such as education level, income category, marital status, and card category.
- CatBoost reduces the need for extensive categorical encoding and can improve model performance when working with customer profile data.
- The model provides strong predictive capability while maintaining robustness against overfitting.

7. Model Evaluation Strategy

Each machine learning model will be evaluated using a comprehensive set of classification performance metrics to ensure reliable assessment of predictive capability.

The evaluation framework includes:

Accuracy

- Measures the overall proportion of correct predictions made by the model.
- Provides a general performance overview but will be interpreted alongside other metrics due to potential class imbalance in churn datasets.

Precision

- Measures the proportion of customers predicted as churners who actually churn.
- Important for evaluating the effectiveness of retention campaigns and avoiding unnecessary customer interventions.

Recall

- Measures the ability of the model to correctly identify customers who are likely to churn.
- Particularly important for PayStone because failing to identify high-risk customers may result in lost revenue opportunities.

F1-Score

- Provides a balance between precision and recall.



- Useful when the churn dataset contains an imbalance between retained and churned customers.

ROC-AUC Score

- Measures the model's ability to distinguish between customers who churn and customers who remain active across different classification thresholds.
- A higher ROC-AUC score indicates stronger predictive discrimination.

Confusion Matrix

- Provides a detailed breakdown of correct and incorrect predictions.
- Helps identify false positives and false negatives associated with churn predictions.

Precision-Recall Curve

- Used to evaluate model performance when dealing with potential class imbalance.
- Provides insights into the trade-off between identifying churn customers and avoiding incorrect predictions.

Stratified K-Fold Cross-Validation

- Used to evaluate model stability and ensure performance consistency across different subsets of customer data.
- Maintains the proportion of churn and non-churn customers within each validation fold.

Hyperparameter Optimisation

To improve model performance, hyperparameter optimisation will be performed using:

GridSearchCV

- Systematically evaluates predefined combinations of model parameters.
- Identifies the optimal parameter configuration that maximises predictive performance.

RandomizedSearchCV

- Efficiently explores a wider range of hyperparameter combinations.
- Reduces computational cost while identifying strong-performing model configurations.

The final production model will be selected based on predictive performance, robustness, interpretability, computational efficiency, and suitability for deployment within PayStone's customer retention framework.

8. Explainable Artificial Intelligence (XAI)

To ensure transparency, accountability, and responsible AI adoption, the selected machine learning model will be interpreted using **SHAP (SHapley Additive Explanations)**.

SHAP will provide:

Global Feature Importance



- Identifies the most influential factors affecting customer churn across the entire customer population.
- Helps PayStone understand major churn drivers, such as declining transaction activity, inactivity periods, or reduced product engagement.

Individual Customer Prediction Explanations

- Provides customer-level explanations showing why a specific customer has been classified as high churn risk.
- Enables targeted retention actions based on individual customer circumstances.

Identification of Key Churn Drivers

- Highlights behavioural, financial, and demographic factors contributing to customer attrition.
- Supports the development of personalised retention strategies.

Improved Trust and Transparency

- Enables business users, risk teams, and decision-makers to understand and validate AI-generated predictions.
- Supports responsible AI practices and improves confidence in machine learning adoption.

This explainability framework allows PayStone to move beyond prediction by understanding **why customers churn** and developing targeted interventions to improve customer retention.

9. Technology Stack

Programming Language

Python

- Used as the primary programming language for data analysis, exploratory data analysis, feature engineering, machine learning development, model evaluation, and API implementation.
- Provides access to a comprehensive ecosystem of data science and artificial intelligence libraries.

Data Processing and Feature Engineering

Pandas

- Used for loading, cleaning, transforming, and analysing customer datasets.
- Supports data preparation activities required before machine learning modelling.

NumPy

- Used for numerical computation, mathematical operations, and efficient handling of structured data arrays.



Data Visualisation

Matplotlib

- Used for creating statistical charts, model evaluation plots, and exploratory data analysis visualisations.

Seaborn

- Used for advanced statistical visualisations such as correlation analysis, feature distributions, and customer behaviour analysis.

Plotly

- Used to create interactive visualisations that allow deeper exploration of customer churn patterns, model performance, feature relationships, and prediction outcomes.

Machine Learning Framework

Scikit-learn

- Used for data preprocessing, feature transformation, model training, performance evaluation, and machine learning workflow management.

Machine Learning Models

- **Logistic Regression** – Provides an interpretable baseline model for churn prediction.
- **Decision Tree** – Captures non-linear relationships and provides explainable decision rules.
- **Random Forest** – Improves prediction stability through ensemble learning.
- **XGBoost** – Provides high-performance gradient boosting predictions.
- **LightGBM** – Enables efficient and scalable gradient boosting modelling.
- **CatBoost** – Handles categorical customer variables effectively.

Model Optimisation and Validation

- **GridSearchCV** – Performs systematic hyperparameter tuning.
- **RandomizedSearchCV** – Efficiently searches parameter combinations.
- **Stratified K-Fold Cross-Validation** – Ensures reliable model evaluation and generalisation.

Explainable AI

SHAP (SHapley Additive Explanations)

- Used to interpret model predictions and identify the factors influencing customer churn.



- Supports transparent AI decision-making by providing both global and customer-level explanations.

Model Deployment

Joblib

- Used to save and load trained machine learning models for deployment.

FastAPI

- Used to develop a lightweight REST API capable of delivering real-time churn predictions.

Render

- Used as the cloud deployment platform for hosting the prediction API and making the model accessible for applications.

Business Intelligence

Microsoft Power BI

- Used to develop an interactive customer churn monitoring dashboard.
- Provides business stakeholders with insights into churn trends, customer risk segments, prediction results, model performance, and key churn drivers.

Development Environment

Jupyter Notebook

- Used for experimentation, exploratory analysis, model development, and documentation.

Visual Studio Code

- Used for software development, API creation, testing, and project management.

Git & GitHub

- Used for version control, source code management, collaboration, and project documentation.

AI Productivity Tools

ChatGPT (OpenAI)

- Used for technical research, coding support, documentation, and problem-solving.

Claude AI

- Used for analytical assistance, technical exploration, and documentation improvement.

GitHub Copilot

- Used for AI-assisted programming and development productivity.



Microsoft Copilot

- Used for productivity enhancement and business intelligence workflows.

Google Gemini

- Used for AI-assisted research, technical exploration, and analytical support.

Descriptions of Variables

No. Variable	Data Type	Full Meaning / Description	Business Interpretation
1 Churn	Categorical	Customer churn/attrition status	Indicates whether the customer is an Existing Customer or Attrited Customer. This is the target variable.
2 Customer_Age	Numerical	Age of the customer in years	Captures demographic differences in customer financial behaviour.
3 Gender	Categorical	Gender of the customer	Represents a demographic characteristic that may be associated with customer behaviour.
4 Dependent_count	Numerical	Number of people financially dependent on the customer	Provides an indication of household and financial responsibilities.
5 Education_Level	Categorical	Highest educational level attained	Represents the customer's educational background.
6 Marital_Status	Categorical	Marital status of the customer	Identifies whether the customer is married,

No. Variable	Data Type	Full Meaning / Description	Business Interpretation
7 Income_Category	Categorical / Ordinal	Customer annual income band	single, divorced, or has an unknown status. Represents the customer's income range and provides an ordered measure of income level.
8 Card_Category	Categorical / Ordinal	Type or tier of credit card held	Represents the card product level: Blue → Silver → Gold → Platinum.
9 Months_on_book	Numerical	Number of months the customer has been with the institution	Measures the duration of the customer's relationship with the institution.
10 Inactive_Months_Last_12_Months	Numerical	Number of inactive months during the previous 12 months	Measures customer inactivity and potential disengagement.
11 Contacts_Last_12_Months	Numerical	Number of customer contacts during the previous 12 months	Measures customer interaction with the institution.
12 Credit_Limit	Numerical	Maximum credit available to the customer	Represents the customer's approved borrowing capacity.
13 Total_Revolving_Bal	Numerical	Total revolving credit balance	Measures the outstanding revolving credit balance.
14 Avg_Open_To_Buy	Numerical	Average unused credit available	Measures the amount of credit remaining after utilisation.

No. Variable	Data Type	Full Meaning / Description	Business Interpretation
15 Transaction_Amount_Change_Q4_Q1	Numerical	Change in transaction amount between Q4 and Q1	Measures changes in customer spending behaviour between the two periods.
16 Total_Trans_Amt	Numerical	Total transaction amount	Measures the total monetary value of customer transactions.
17 Total_Transaction_Count	Numerical	Total number of transactions	Measures customer transaction frequency and product engagement.
18 Transaction_Count_Change_Q4_Q1	Numerical	Change in transaction count between Q4 and Q1	Measures changes in customer transaction frequency.
19 Avg_Utilization_Ratio	Numerical	Average credit utilisation ratio	Measures the proportion of available credit being utilised by the customer.

The logo graphic at the top of the page consists of several horizontal bars in teal and dark grey. On the right side, there is a stylized 'A' symbol inside a teal circle.

AMDARI



Amdari Limited UK

128, City Road, London, EC1V 2NX, UNITED KINGDOM